

PEER-TO-PEER (P2P) DATASET FEDERATION

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P2P dataset federation

- PoC
- Motivation use case: sharing IQM produced my MRIQC, across institutions without centralized infrastructure
- Not to move MRIQC web API
- Jupyter notebook can open and query a federated collection of Parquet files distributed across an ad-hoc peer-to-peer (P2P) network

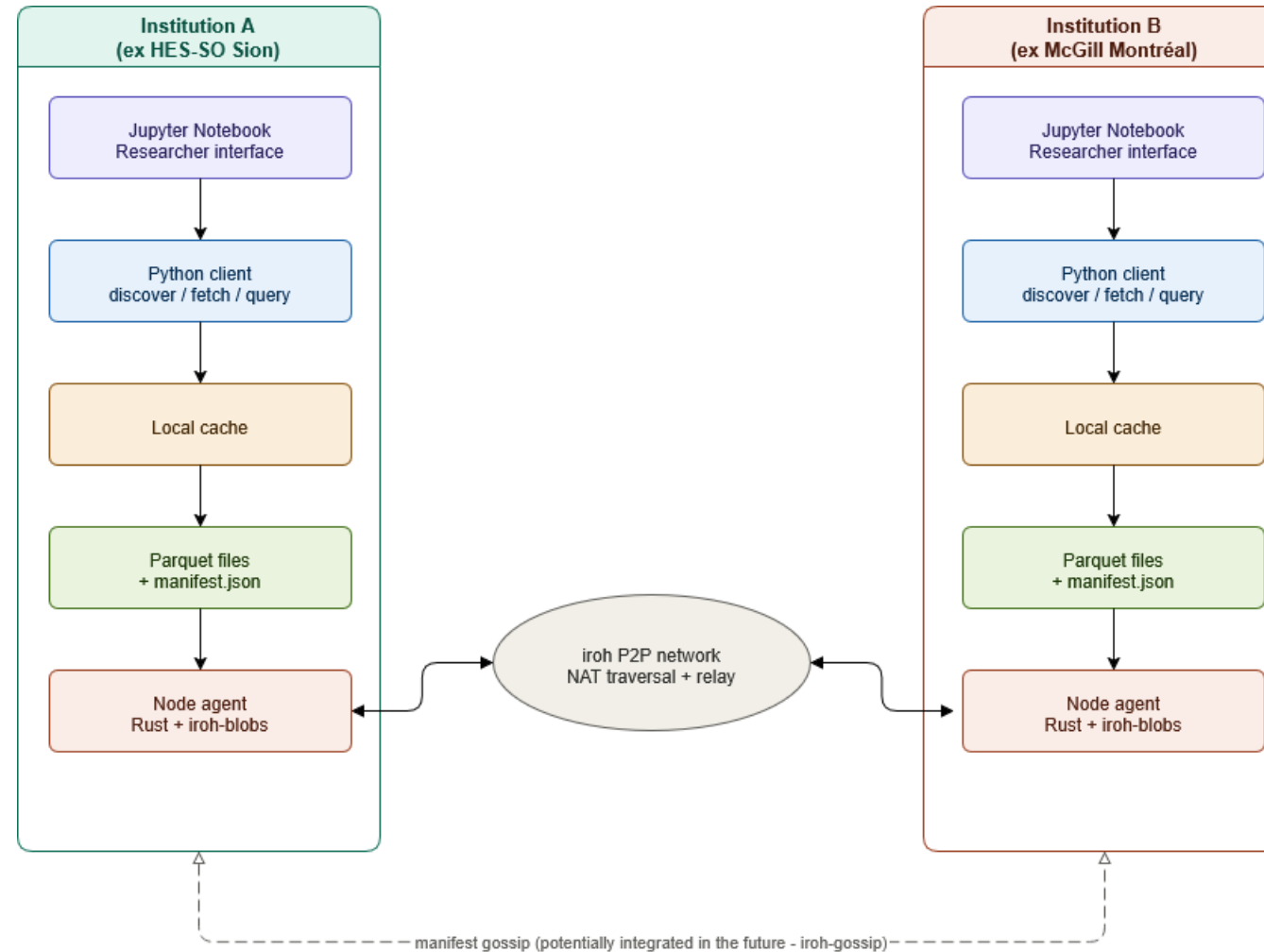
Problem

- MRIQC IQMs – Different place
- Analyze them together from a Jupyter notebook without any shared infrastructure
- Comparison:

	Central server (Web-API)	P2P
Best for	Public sharing	Closed consortiums, no infra available
Data	Uploaded to a third-party server	Stays at each institution
Setup	Simple for the user	More complex, but autonomous
Dependency	Server + funding	None

System architecture

- First draft



Limit of the PoC

Donnée du travail de bachelor

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Study programme	Academic year	Thesis reference
ISC	2025-26	ISC-ID-26-6
Principal	Expert	Place of execution
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Confidential work	Professor	Co-supervisor
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Date version	May 12, 2026 - v1.30	

Title	Peer-to-Peer (P2P) Dataset Federation
Description	This TB will deliver a proof-of-concept demonstrating that a Jupyter notebook can open and query a federated collection of Parquet files distributed across an ad-hoc peer-to-peer (P2P) network of 2–5 machines, with access that appears local and seamless to the user. The P2P transport layer is provided by iroh, an existing Rust-based networking toolkit; the student's implementation work focuses on the Python client layer that discovers, fetches, caches, and exposes distributed Parquet files to standard dataframe tools (PyArrow, pandas, DuckDB). The motivating use case is the sharing of Image Quality Metrics produced by MRIQC, a neuroimaging quality control tool, across institutions without centralized infrastructure. The project prioritizes a working end-to-end demonstration over low-level protocol development.
Objectives	Knowledge, skills, and competences to be acquired: - Understand the fundamentals of peer-to-peer networking (peer discovery, data integrity, NAT traversal) and configure an ad-hoc iroh network using existing iroh tooling. - Design and implement a Python client library that discovers available Parquet files on the network, fetches missing files into a local cache, and presents the distributed collection as a single dataset accessible from a Jupyter notebook. Validate application logic on a containerized testbed (Docker Compose). - Deliver a reproducible demonstration package: a working Jupyter notebook, setup scripts, dataset conventions, and a short evaluation report documenting performance, limitations, and recommended next steps. <i>Bonus:</i> Validate the prototype on 2–5 physical machines under realistic network conditions (peer join/leave, partial availability, NAT traversal).

Addendum — Bachelor thesis description

Project type	☐ Exploratory ☒ Implementation
Data dependencies	☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 1 No data 2 Data collected, cleaned and ready to be used 3 Data collected, cleaning needed 4 Data not collected but available online 5 Data not collected and dependent on external actors

Initial goal

- Demonstrate the feasibility of sharing MRIQC's IQMs between peers without a central server
- Milestone
 1. Onboarding & Technical Design (W1-W2)
 2. Core P2P sharing prototype (W3-W6)
 3. Python/Jupyter integration and connectivity validation (W7-W10)
 4. Robustness and evaluation in a 2-5 machine testbed (W11-W13)
 5. Packaging, documentation, and final demonstration (W14)
- Optional
 - Parquet partial-read optimization
 - Iroh-gossip for discovery

Who might find this interesting?

- Consortium of 4 hospitals wanting to compare IQMs
- A lab wanting to aggregate IQMs from its own distributed machines
- Situation where the central server is unreachable